WHEN CONFIDENCE IS NOT CORRECTION: A CONTROLLED NEGATIVE STUDY OF TRAINING-FREE POST-HOC ADAPTATION FOR RGB SINGLE-OBJECT TRACKING

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Index Terms— Visual object tracking, template memory, autoregressive tracking, negative results, reproducibility.

ABSTRACT

Training-free interventions commonly use confidence to decide whether to alter a trackerś template memory, state, search region, or decoded box. We study a narrow question: for a released frozen FARTrackSparse checkpoint, does a failure-predictive signal provide enough information to choose a better post-hoc action? The development protocol uses public GOT-10k validation (180 sequences, 21,007 frames), while LaSOT Testing remains held out. Four complete development evaluations show AO changes of -0.00823, -0.02000, -0.03204, and -0.00273 for transactional memory, counterfactual template gating, stable-anchor recovery, and logarithmic sampling. A passive coordinate-entropy probe predicts low-IoU frames (AUROC 0.9067), but split-controlled memory, motion, search, posterior, token, and flow actions fail their pre-specified effect or efficiency gates. We formalize the gap between predicting failure and predicting a positive conditional action advantage, and report the negative evidence with its evaluation tier, uncertainty, and artifact-retention limits. We make no claim that post-hoc adaptation is universally ineffective. A baseline-only LaSOT reference gives success AUC 0.6148 on 280 Protocol-II sequences; no rejected candidate was run on that held-out set.

1. INTRODUCTION

Autoregressive tracking exposes a practical tension: recent templates improve adaptation, while an erroneous crop may contaminate later predictions. FARTrack uses autoregression and inter-frame sparsification to obtain fast tracking [1]. We asked a deliberately narrow question: can untrained control of its inference-time template memory improve a released, frozen FARTrackSparse checkpoint without additional training?

This paper reports a bounded answer: the tested mechanisms did not meet their pre-specified deployment criteria. We regard this as useful evidence because a plausible protection can silently disrupt the update distribution expected by a frozen tracker. We do not replace this negative result with a post-hoc positive claim, and do not claim that FARTrackSparse is optimal or that other trackers cannot benefit from adaptation.

2. FROM RISK TO ACTION ADVANTAGE

Let Z_t contain the information already available to the frozen tracker at frame t: the current crop, template memory, logits, decoded box, and causal box history. Let $a_0(Z_t)$ be the released action and $a_g(Z_t)$ a training-free intervention. The quantity needed for a beneficial intervention is its conditional action advantage

$$A_g(Z_t) = \mathbb{E}[\mathrm{IoU}(a_g(Z_t), B_t^*) - \mathrm{IoU}(a_0(Z_t), B_t^*) \mid Z_t]. \quad (1)$$

A reliability score can rank a low-IoU event, denoted $Y_t = 1$, without determining the sign of A_g. In particular, a threshold policy that replaces a_0 only when $r(Z_t) > \tau$ has mean gain

$$\mathbb{E}[\mathbf{1}\{r(Z_t) > \tau\} A_g(Z_t)], \quad (2)$$

which is not implied by a high AUROC for r. This distinction is the central testable premise of our study.

The same limitation appears in several common controls. Locally, if $u(a; Z_t) = \mathbb{E}[\mathrm{IoU}(a, B_t^*) \mid Z_t]$ is stationary at a_0 with negative-semidefinite Hessian H, then a small correction δ has $u(a_0 + \delta; Z_t) - u(a_0; Z_t) \approx \frac{1}{2}\delta^\top H \delta \leq 0$ unless it exploits a signed residual. For a random-walk appearance model, an old template accumulates a stale-state term proportional to elapsed time, whereas a current template can be contaminated; a confidence alarm alone does not identify which error is smaller. Likewise, a motion mixture helps only when its residual is stably anti-correlated with the visual residual. Coordinate-wise posterior changes need not improve the joint four-boundary IoU. Finally, a deterministic mapping of Z_t cannot create new information about B_t^*; this data-processing observation does not prove that a_0 is Bayes-optimal, but motivates testing signed action advantage rather than confidence alone.

3. PROTOCOL

Model and invariants. Every experiment uses the released FARTrackSparse epoch-15 checkpoint, the original prediction decoder, the original result writer, and the same initialization boxes. Its immutable baseline entry point is documented with the artifacts; the upstream inference implementation is not modified.

Development/test separation. $\mathrm{B}_{\mathrm{dev}}$ is the public GOT-10k validation split. GOT-10k was introduced as a high-diversity generic tracking benchmark with an explicit one-shot protocol [2]. The development metric is mean frame IoU (AO) over 180 sequences and 21,007 frames, calculated by a read-only local script which checks sequence coverage and prediction lengths. Full official LaSOT Testing is $\mathrm{B}_{\mathrm{test}}$. LaSOT supplies dense, long sequences for SOT evaluation [3]. It was checksum-verified and not accessed until all candidate decisions were frozen. The $\mathrm{B}_{\mathrm{test}}$ run is baseline-only because no candidate survived $\mathrm{B}_{\mathrm{dev}}$. It measured success AUC 0.6148, precision at 20 pixels 0.6446, and normalized precision AUC 0.6399. This is an external baseline reference, not a new-method result; no cross-run comparison to upstream repository tables is claimed.

Statistics and speed. We report the mean of per-frame IoUs. For a candidate-baseline comparison, we resample the 180 per-sequence AO differences 10,000 times with seed 20260819 and report a percentile 95% interval. Aggregate FPS is total frames divided by the sum of saved per-frame times. These intervals describe the observed development set; they are not independent-video population inference.

4. INTERVENTIONS

1. **TRM-FAR**: a transaction ledger holds writes under three causal anomaly signals and rolls back after sustained failure.
2. **Counterfactual agreement**: old-only and recent-only template views gate a write when their coordinate predictions disagree. It requires two extra forwards after warm-up.
3. **Stable-anchor recovery**: persistent failures temporarily replay the initial template and the last accepted view.
4. **Recent-consensus rebinding**: all frames are written, but fixed template slots are rebound using untrained RGB descriptor medoids.
5. **Appearance-diverse rebinding**: all frames are written, while slots retain chronological max–min RGB descriptor exemplars.
6. **Logarithmic sampler**: the upstream sampler is selected without adding a model forward or changing the write/mask implementation.

All implementations are separate tracker entry points. Focused unit tests and GPU smoke tests preceded evaluation. Counterfactual agreement initially exposed an out-of-range indexing defect after a rejected write; it was repaired by indexing its bounded template pool by accepted writes, and only the repaired version was evaluated.

Reliability follow-up. A passive probe measured coordinate-bin entropy, branch disagreement, and IoU without affecting outputs. Entropy identified frames with IoU below 0.2 with AUROC 0.9067, but a signal that predicts failure need not prescribe its repair. We therefore pre-registered a calibration/selection/confirmation partition by sequence identifier. The following actions were stopped before B_{test}: dual-lane memory, causal state mixing, delayed search expansion, rare two-view arbitration, entropy-age recent-window reading, and posterior-shape adaptive point projection. Their broad motivation overlaps uncertainty-aware tracking literature [4, 5, 6]; this study does not claim conceptual novelty for uncertainty gating.

Finally, we audited token-level rather than template-level control. The native FARTrack masks retain 37, 25, 13, or 5 of 49 template tokens. Even a rare, disagreement-triggered stronger mask reduced calibration AO by 0.00901. An independent bidirectional sparse optical-flow arbitration was also rejected on two fixed screens: it reduced AO where it changed boxes and cost 27.94% FPS on the triggered sequence. Optical-flow tracking is established prior art [7]; it is reported only as a feasibility ablation.

5. RESULTS

Table 1. Complete B_{dev} results. AO is mean frame IoU.

Method	AO	Δ AO	95% paired interval	FPS
Immutable FARTrackSparse	0.832741	0.000000	–	40.07
TRM-FAR	0.822471	-0.008228	[-0.019499, 0.004104]	42.16
Counterfactual agreement	0.806470	-0.019995	[-0.037073, -0.003256]	23.47
Stable-anchor recovery	0.797473	-0.032038	[-0.050311, -0.015034]	40.90
Logarithmic sampler	0.830650	-0.002730	[-0.010716, 0.003984]	31.32

Risk-stratified action advantage. We performed a read-only post-hoc audit of the valid CSPP selection output (not a new policy selection). The q85 entropy and disagreement thresholds were estimated only from the 60 calibration sequences. The 60 held selection sequences provide 7,955 non-initial frames after aligning the byte-identical passive baseline output, reliability logs, CSPP output, and public GT. High-entropy frames have mean CSPP-minus-baseline IoU +0.00941 (95% sequence bootstrap [-0.01732, 0.04072]); high-disagreement frames have +0.00823 ([-0.01641, 0.03306]). Both intervals cross zero. Across the 4,487 frames whose CSPP box changed, the mean is -0.00208 ([-0.02557, 0.02048]). Thus the risk strata have suggestive point estimates but do not establish the signed action advantage required for deployment.

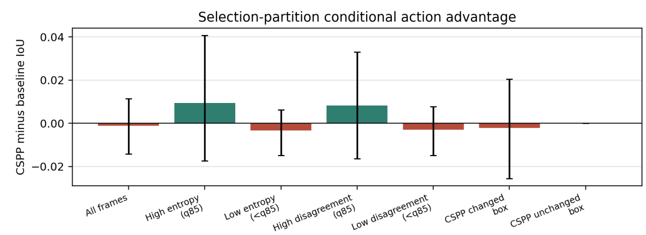

Fig. 1. Read-only CSPP-minus-baseline IoU on the held selection partition. Error bars are 95% sequence-bootstrap intervals; thresholds were fixed on the separate calibration partition. Positive high-risk point estimates remain too uncertain to support a new action policy.

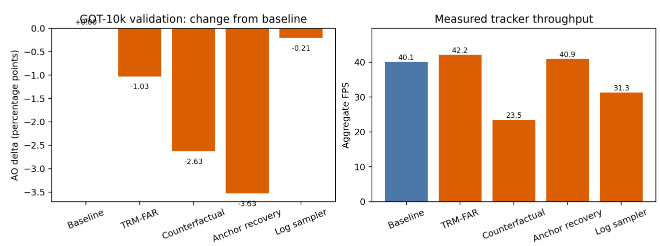

Fig. 2. Measured complete GOT-10k validation outcomes. Orange bars are rejected candidate mechanisms; the graphic reports observed B_{dev} data only and is not a held-out benchmark result.

TRM-FAR improved 70 sequences and degraded 110; stable-anchor recovery improved 66 and degraded 114; counterfactual agreement improved 77 and degraded 103. No candidate in Table 1 was tuned after its complete development result or run on B_{test}.

Effect distributions. A read-only distribution audit of the same 180 per-sequence AO values shows mixed outcomes rather than uniform degradation. Nevertheless, TRM-FAR, counterfactual agreement, and stable-anchor recovery have negative medians (-0.00393, -0.00466, and -0.00726); their five largest per-sequence losses account for 33.76%, 35.26%, and 33.54% of total negative delta. The logarithmic sampler has a near-zero median (-0.00053), but its five largest losses account for 55.89%. Thus the aggregate failures cannot be summarized as a single shared catastrophic case, and positive outcomes on some sequences do not establish a fixed deployment rule.

The two soft alternatives did not pass their prespecified focused-screen gates. Recent-consensus rebinding obtained 0.907830 versus

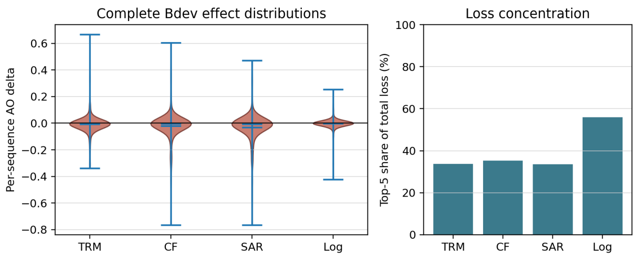

Fig. 3. Read-only distribution audit of the complete B_{dev} results. Left: per-sequence AO deltas. Right: the share of total negative per-sequence delta contributed by each method's five largest losses.

0.909803 for the baseline on a fixed 60-frame screen. Appearance-diverse rebinding obtained a small 0.000838 AO increase on a fixed 80-frame screen, but incurred a 16.57% FPS loss (21.93 versus 26.28 FPS), exceeding the 5% gate. These are screening observations, not generalization estimates, and were not expanded to a full development evaluation.

Table 2. Split-controlled reliability follow-up. Baseline calibration AO is 0.847678. All candidates were rejected before held-out testing.

Action	Δ AO	FPS change	Stop point
Dual-lane memory	-0.003999	-15.6%	selection
State mixing	+0.000644	-58.6%	selection
Delayed search	-0.003339	+18.3%	selection
Two-view arbitration	-0.000046	-11.6%	calibration
Recent-window read	-0.000685	+10.8%	calibration
Posterior projection	-0.009987	–	calibration
Token quarantine	-0.009010	-1.1%	calibration
Sparse-flow arbitration	-0.002827	-27.9%	screen

6. DISCUSSION AND LIMITATIONS

The shared pattern is that hard intervention can sacrifice normal appearance adaptation more often than it prevents memory pollution. In this checkpoint, the tested uncertainty, motion-inconsistency, and disagreement signals did not establish a reproducible positive action advantage for their paired controls. Further, changing the effective template distribution may create an input regime unseen during training. For the closest-to-neutral logarithmic sampler, sequence-level AO changes have weak Spearman associations with public GOT-10k labels and simple geometric summaries (absolute $\rho \leq 0.102$ for absence, cover-label mean, cut fraction, relative motion, scale variation, and length). Thus no single observed attribute identifies when the fixed rule helps. This study does not establish that template control is never useful; it only rejects these implementations and fixed settings on the stated protocol.

Limitations are substantial. There is one released checkpoint, one full public development split, RGB descriptors rather than learned confidence, and no positive method. The LaSOT outcome is a baseline-only reference; it cannot validate a rejected candidate. GOT-10k test server submission has not been performed.

7. REPRODUCIBILITY AND DISCLOSURE

All source changes, commands, structured metrics, logs, and immutable checkpoint path are documented in the accompanying experiment log. Data archives and outputs reside on the data disk. This draft was prepared with AI assistance; human authors must verify all text, select authorship, and approve any submission. No funding, conflict-of-interest, or authorship statements are asserted because they were not provided.

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